AI-POWERED RECOMMENDATION ENGINE

CardWise

Product Requirement Document

User Stories · Prioritization Matrix · Architecture

Axis Bank × FreechargeBiz

CW

May 2026 · Version 1.0 · Confidential

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01 Executive Summary

■ CardWise is an AI-powered credit card recommendation engine built for FreechargeBiz. It replaces static form-based card finders with a 90-second, conversational, context-aware journey that analyses a user's financial footprint — SMS transactions, soft bureau pull, and explicit preferences — to surface the single best-fit Axis Bank or Freecharge card with a transparent match score.

Product Name	CardWise — AI Credit Card Recommendation Engine
Sponsor	FreechargeBiz (Axis Bank Digital Unit)
Version	1.0 — May 2026
Objective	Increase card application conversion by 40% through hyper-personalized matching
Target Launch	Q3 2026 (MVP in 8 weeks)
Key Metric	Recommendation-to-Application rate ≥ 35%

Strategic Rationale

India's credit card market is growing at 25% YoY, yet conversion rates on card aggregator platforms sit below 8%. The core problem is misalignment: users don't know which card serves them best, and legacy questionnaires are too long, too generic, and produce recommendations that feel like guesses. CardWise solves this with real spend data and a transparent AI match score — building trust and driving decisions.

02 Problem Statement & User Personas

The Fractured Marketplace Journey

When a user visits a credit card comparison platform today, they are met with a wall of cards, sorted by commission rather than fit. If they attempt a "find my card" flow, they face 12–18 static questions, receive a generic result ("Best for Travel!"), and feel no confidence the match is personalised to them. Drop-off rates at the questionnaire stage exceed 70%.

Pain Point	Impact	Magnitude
Too many cards, no clear differentiator	Decision paralysis	HIGH
Long, boring questionnaires (12–18 steps)	70%+ drop-off at step 3	CRITICAL
Recommendations feel generic / un-trusted	Low application intent	HIGH
No context from actual spending behaviour	Poor reward fit post-issuance	HIGH
Hard credit pull fears	Users abandon before consent	MEDIUM

User Personas

Priya, 28 — The Digital Professional

Background: Salaried software engineer, ■1.2L/month in-hand, Bangalore. Spends ■25K+/month on travel, Swiggy, and Amazon. Has one basic HDFC card, wants to upgrade.

Goals: Lounge access, travel insurance, reward points she can actually use

Frustrations: Doesn't have time for long forms. Doesn't trust "best card" lists that feel paid.

Data comfort: Comfortable sharing SMS data and soft bureau pull

> Jobs to be done: ""Show me the one card I should switch to — fast.""

Rajan, 42 — The Family CFO

Background: Self-employed business owner, ■3L+/month turnover, Delhi. Spends heavily on fuel, insurance premiums, and utility bills.

Goals: Maximum cashback on fuel and insurance, high credit limit, reward flexibility

Frustrations: Keeps getting recommended travel cards despite never flying. Complex reward programs he never redeems.

Data comfort: Will share bank statement PDF for a better match

> Jobs to be done: ""Give me real money back on what I already spend.""

Mehak, 22 — The First-Timer

Background: Recent graduate, new job at ■35K/month. No credit history. Heavy UPI user; wants to build credit.

Goals: Zero annual fee, simple cashback, eligibility confidence before applying

Frustrations: Scared of rejection. Doesn't understand reward points vs cashback. No bureau file.

Data comfort: Only comfortable answering questions; no document sharing

> Jobs to be done: ""Just tell me which card I'll actually get approved for.""

03 Core Features & User Journey

End-to-End Flow (5 Screens)

Screen	Name	Key Action	Drop-off Lever
0	Hero + Consent	User selects data sources; reads privacy promise	Social proof, no hard pull badge
1	Personal Info	Name, age, city, employment type	Progress bar, auto-fill from SMS
2	Spending Profile	Category tiles + monthly spend slider	Visual card, max 6 taps
3	Reward Goal	One primary goal + credit score range	Single-select, instant feel
4	AI Analysis	Animated loading with live step indicators	Reduces anxiety, builds trust
5	Recommendation	Card visual, match score, benefits, alt options, CTA	Transparent reasoning, apply CTA

Core Features

Contextual Consent & Data Onboarding

Users select their comfort level across 4 data source tiers: SMS spend analysis, bureau soft-pull, bank statement upload, and explicit answers. Each tier unlocks better match precision. Consent is granular, revocable, and displayed in plain language. RBI AA framework compliant.

Spend Category Intelligence

SMS parser extracts merchant-category spend data from bank alerts (HDFC, SBI, ICICI, Kotak, Axis) using regex + ML classification into 12 categories: travel, fuel, dining, shopping, utilities, insurance, EMI, healthcare, education, subscriptions, entertainment, cash. No PII stored.

AI Recommendation Engine

A weighted scoring model evaluates 23 parameters across 8 eligible cards. Parameters include: monthly spend volume, category distribution, credit score band, employment type, city tier, existing cards held, age, and goal preference. Each card receives a match score (0–100) per user.

Transparent Match Score

The recommendation page shows a breakdown of the 94% match across four dimensions: Reward Alignment, Spend Category Match, Eligibility Confidence, and Fee-to-Benefit Ratio. This is the anti-black-box feature that builds trust and reduces decision anxiety.

Dynamic Card Visualisation

A rendered credit card graphic (matching the actual card art) is shown with the user's name on it from the first personalisation step — creating an emotional connection and ownership feeling before they even apply.

"Also Consider" Rail

Two to three alternative cards are displayed with a one-line differentiation. Users can explore alternatives without losing their primary recommendation context.

Instant Apply CTA

The apply button triggers an in-app pre-filled application form. For users who granted bureau consent, PAN and eligibility data pre-populates the form — reducing keystrokes by 60%.

04 Technical Architecture & Data Strategy

System Architecture Overview

CardWise is built as a microservices architecture deployed on AWS. The recommendation engine runs as a separate inference service to allow independent scaling during traffic peaks (e.g. salary week).

Layer	Component	Technology	Purpose
Frontend	Progressive Web App	React + TypeScript	Consent, questionnaire, result screens
API Gateway	REST + WebSocket	AWS API Gateway	Route requests, auth tokens
Services	SMS Parser Service	Python + spaCy + regex	Extract merchant & amount from bank SMS
Services	Bureau Connector	CIBIL / Experian AA API	Soft-pull credit score, no footprint
Services	Recommendation Engine	Python + scikit-learn	Weighted scoring & card ranking
Services	Profile Service	Node.js	Temp session state, no persistent PII
Data	Card Catalogue DB	PostgreSQL (RDS)	Card features, rewards, eligibility rules
Data	Analytics Store	Redshift	Anonymised funnel & A/B data
Infra	Auth & Consent	AWS Cognito + DynamoDB	Consent ledger, RBI AA compliance

Data Inputs & Recommendation Parameters

Tier 1: SMS Transaction Data (Primary Signal)

When a user grants SMS permission, the app reads the last 90 days of bank alert SMSes. A parser identifies: merchant name, amount, category (via merchant-category mapping), frequency, and payment mode. Output: spend vector [category → monthly ■ amount]. This data never leaves the device in raw form — only the aggregated spend vector is sent to the backend.

Tier 2: Bureau Soft-Pull (Eligibility Signal)

Via Account Aggregator (AA) framework with user consent, CardWise retrieves: CIBIL score band (not raw score), active credit lines count, and utilisation ratio. No hard enquiry is logged. This data gates eligibility — users with score < 650 are routed to secured/entry cards.

Tier 3: Explicit Questionnaire (Context Signal)

Four screens capture: employment type, age bracket, primary spend categories (multi-select), monthly spend estimate (slider), and reward goal (single-select). These are always collected, forming the minimum viable input for a recommendation.

Recommendation Algorithm

Parameter	Weight	Source
Monthly spend volume	20%	SMS + Slider
Category spend distribution	25%	SMS + Tile selection
Reward goal alignment	20%	Questionnaire
Credit score band	15%	Bureau / Self-reported
Employment type	10%	Questionnaire
Fee-to-benefit ratio	10%	Card catalogue

■ Privacy by Design: All PII (name, mobile) is stored only in encrypted session tokens (15-min TTL). Aggregated spend vectors are anonymised before any analytics logging. No user data is shared with Axis Bank without an explicit application action from the user.

05 User Stories

Stories are grouped by theme and follow the format: As a [User], I want to [Action], so that [Value]. Acceptance criteria use Gherkin-style Given/When/Then notation.

Onboarding & Consent

US-01

As a new user, I want to understand what data is being accessed and why, so that I can make an informed consent decision without feeling surveilled.

Acceptance Criteria:

• Given I land on the app, when I see the consent screen, then each data source shows a plain-language description and benefit.
• Given I toggle off SMS access, when I proceed, then the recommendation uses only questionnaire data with a lower accuracy badge.
• Given I consent, when I change my mind mid-flow, then I can withdraw consent and the session data is deleted.

US-02

As a privacy-conscious user, I want assurance that no hard credit pull is triggered, so that my credit score is not affected by exploration.

Acceptance Criteria:

• Given I select bureau data, when the soft-pull runs, then no enquiry footprint is created.
• Given bureau consent is granted, when results load, then a "No credit impact" badge is displayed.

Data Collection & Personalisation

US-03

As a user who has granted SMS access, I want my spending categories to be auto-detected, so that I don't have to manually enter what I already spend on.

Acceptance Criteria:

• Given SMS permission is granted, when the parser runs, then spend categories are pre-selected on screen 2.
• Given categories are auto-detected, when I disagree, then I can manually add or remove categories.
• Given SMS parsing completes, when I see the slider, then the default amount reflects my 90-day average.

US-04

As a user without SMS access granted, I want a quick and visual questionnaire, so that I can complete the flow in under 90 seconds.

Acceptance Criteria:

• Given no SMS access, when I reach screen 2, then tile cards show spend categories for manual selection.
• Given I select categories, when I adjust the spend slider, then the value updates in real-time in Indian numeral format.
• Given screen 3 loads, when I select one reward goal, then the selection is confirmed with a visual highlight.

Recommendation Engine

US 05

As a user who has completed the flow, I want to see a single recommended card with a clear match percentage, so that I feel confident the recommendation is personalised to me.

Acceptance Criteria:

• Given I complete screen 3, when the AI analyses my profile, then exactly one primary card is shown.
• Given the result loads, when I see the match score, then a breakdown of 4 contributing dimensions is visible.
• Given I have travel spend > 30% of total, when the engine runs, then a travel-category card is recommended.

US 06

As a user with a credit score below 650, I want to be shown an eligible entry-level card, so that I am not recommended a card I will be rejected for.

Acceptance Criteria:

• Given credit score < 650, when the engine runs, then only secured or entry-level cards are in the candidate set.
• Given eligibility confidence < 70%, when the result is shown, then a "Pre-qualify check" CTA replaces the direct Apply button.

US 07

As a user, I want to see 2–3 alternative card suggestions below my primary recommendation, so that I can explore options without starting over.

Acceptance Criteria:

• Given the primary card is shown, when I scroll down, then 2–3 alternative cards appear with a single differentiator each.
• Given I click an alternative, when the detail opens, then match score and key benefits are shown.

Application & Conversion

US 08

As a user ready to apply, I want a pre-filled application form, so that I can complete the process in under 2 minutes.

Acceptance Criteria:

• Given bureau consent was granted, when I tap Apply, then PAN and address fields are pre-populated.
• Given I tap Apply, when the form loads, then my name and mobile from session are pre-filled.
• Given I submit the form, when Axis approves, then I receive an in-app approval with virtual card within 2 minutes.

US 09

As a user not ready to apply, I want to save my recommendation and return later, so that I don't lose my match results.

Acceptance Criteria:

• Given I am on the result screen, when I tap "Save for later", then a unique link is generated.
• Given the link is opened within 7 days, when the page loads, then the same recommendation is shown.

06 Prioritisation Matrix (MoSCoW)

Stories and features are classified using MoSCoW: Must Have (MVP), Should Have (v1.1), Could Have (v2.0), and Won't Have (out of scope). RICE score = (Reach × Impact × Confidence) / Effort.

ID	Feature	MoSCoW	RICE Score	Sprint
US-01	Granular consent & data source selection	Must	324	1
US-02	Soft bureau pull with no credit impact	Must	288	1
US-03	SMS spend auto-detection	Must	360	1-2
US-04	Visual questionnaire (< 90 sec flow)	Must	400	1
US-05	AI recommendation with match score	Must	450	2
US-06	Eligibility gating by credit score	Must	312	2
US-07	"Also consider" alternative card rail	Should	210	3
US-08	Pre-filled application form	Should	280	3
US-09	Save recommendation link	Could	140	4
—	Bank statement PDF upload + parsing	Could	168	4
—	WhatsApp notification of saved result	Could	120	5
—	Comparison mode (2 cards side by side)	Should	196	4
—	Chatbot clarification ("What is a lounge?")	Won't	—	Backlog
—	Open Banking real-time transaction feed	Won't	—	v3.0

MVP Scope (Sprint 1–2)

MVP includes US-01 through US-06. This delivers the complete end-to-end flow: consent → questionnaire → SMS parsing → AI recommendation with match score → direct apply CTA. Target: 8 weeks to production, 4-person squad (1 PM, 2 FE, 1 BE, 1 Data Scientist).

07 Success Metrics & KPIs

Metric	Baseline	MVP Target	v2 Target
Flow completion rate	28%	55%	70%
Recommendation-to-Apply rate	8%	35%	45%
Avg. time in flow	6.5 min	< 90 sec	< 75 sec
SMS consent grant rate	—	> 60%	> 72%
Card-fit satisfaction (post 30d)	—	CSAT ≥ 4.2/5	CSAT ≥ 4.5/5
Monthly card issuances via platform	Baseline	+40%	+100%

08 Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
SMS permission denied on iOS (no API)	HIGH	HIGH	Fallback to questionnaire; no feature loss. Show accuracy trade-off badge.
Bureau API latency > 3s	MED	MED	Async call with skeleton loader. Recommendation proceeds without bureau if timeout.
Sparse SMS data (new SIM / UPI-only users)	MED	HIGH	Detect sparse data early; auto-upgrade to manual questionnaire mode.
Regulatory change to AA framework	LOW	HIGH	Architecture decouples bureau call into swappable adapter. 2-sprint swap time.
Model bias toward high-spend users	MED	MED	Monthly bias audit. Separate recommendation logic for spend < ■ 15K/month.

09 Roadmap

Phase	Timeline	Scope	Milestone
MVP	Wk 1–8	US-01 to US-06 · Consent, SMS parsing, Recommendation engine, Apply CTA	Public Beta Launch
v1.1	Wk 9–14	Alt cards rail · pre-filled apply form · Save & share link · A/B testing framework	Full Conversion Suite
v2.0	Wk 15–22	Bank statement PDF parse · Comparison mode · Personalized reward calculator · Re-targeting push	Retention & upsell
v3.0	Wk 23+	Open Banking real-time feed · Cross-sell (loans, insurance) · Portfolio optimizer for existing cardholders	Platform Expansion

> Submission deadline: May 31, 2026 · Live Prototype: cardwise.freechargbiz.in · Prepared by: Product Team, FreechargeBiz · For stakeholder review only.